FLEET TEAM DIGEST

737MAX-FTD-34-23008

Issue Title : Unexpected Vertical Deviation/Airplane Level Off during VNAV Gradient Leg Descents

Airplane Model	ATA	Minor Model(s)
737MAX	3461-00	-10, -7, -8, -8200, -9, -BBJ
Other Model(s)	**ECCN**	
	9E991	

Originated Date	Last Revised Date	Created On
11/22/2023	11/22/2023	10/18/2023
Estimated Completion Date	**Next Update Date**	

Description

Boeing has received a few operator reports of unexpected level off and jump in the observed FMC calculated path vertical deviation during gradient leg descents (descents using a coded vertical angle). In some events the observed deviation included a Vertical Navigation (VNAV) disconnect.

Background:

GE Simulation and code review identified the following root cause:

The FMC is performing a path re-prediction as a result of an FMC miscompare that can randomly occur when a difference between two FMCs (in a dual FMC installation) is detected. A miscompare can be the result of a configuration change such as flap extension causing a momentary FMC disagreement and subsequent FMC re-sync. The miscompare is addressed quickly and does not trigger the FMC Disagree message logic. However, as a result of the FMC miscompare and resync, the FMC can perform a path re-prediction.

If the previous waypoint has an "At or Above" altitude constraint; that "At or Above" altitude is used by the FMC path re-prediction calculation. This will result in the airplane being below the re-predicted FMC path and subsequent momentary level off until the new path is intercepted.

It should be noted the FMC will continue to comply with all flight plan altitude constraints; however, the airplane will have a slightly steeper path.

For this condition to occur, the airplane approach segment must include an AT OR ABOVE history waypoint altitude constraint. A jump in the FMC VDEV is the result of FMC mis-compare that leads to an FMC flight path re-prediction.

The jump in the FMC Vertical Deviation (VDEV) for this condition does not occur when the previous constraint is an AT (not an AT/Above) or if the constraint is removed.

Boeing has determined this condition as Non-Safety. Boeing has reviewed the condition related to the FMC VDEV level-off condition and the condition is recognizable by the flight crew and appropriate procedures are in place.

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This condition may occur with FMS OPS U10.6 through U13 and is also present on the 737NG Fleet (reference 737NG-FTD-34-19003).

Status

The root cause for this condition has been addressed in the U14 FMC software update (reference 737MAX-FTD-34-18005).

Interim Action

In the event the condition is reported, operators can review the specific procedure for the presence of AT/ABOVE altitude constraints.

If AT/ABOVE constraints are identified, operators can contact their NDB supplier to determine what, if any, change can be made within the NDB to replace AT/ABOVE with AT constraints.

Final Action

Incorporate FMC software U14 or later per Boeing's applicable service bulletin.

Milestones

Note: Target dates are based upon currently available status and have risk of future revision.

Service Bulletin Available: 28-Jan-2022 (Complete)

Reference Categories

Reference Type	Reference Number	Revision Number	Reference Date
Fleet Team Digest Article (FTD)	737MAX-FTD-34-18005		
Fleet Team Digest Article (FTD)	737NG-FTD-34-19003		

Related Categories

- Significant Event Report : List of significant in-service or maintenance events reported to Boeing that result in Air turnbacks, ground turnbacks, diversions, rejected takeoffs, in-flight shutdowns, or Significant AOGs. High failure rate of costly/critical systems

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